

Breach Analysis Simulation One

William Pascoe

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Aries Preston

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For this scenario the tenet of the CIA triad that I think applies is integrity. The scenario noted that because the company still had weak SSL encryption the hacker gained access to their web server and installed some files to see if anyone would notice the compromise. This would call into question whether any of the data that is stored on the compromised server has been tampered with and still remains valid. Once the threat has been neutralized the company would probably want to scrub their data to make sure that there was no other malicious activity as a result of the breach.

During the mid-90's people started to have a growing concern for internet security. Because of this Netscape Communications developed SSL, or Secure Socket Layer, it was designed to provide privacy, authentication, and data integrity between web browsers and servers (Thurman, 2026). This was great because things like E-commerce and online banking were becoming more popular. SSL did, however, have some serious vulnerabilities. It was found that hackers could force a downgrade of protocol which allowed them to decrypt ciphertext. This was known as a POODLE attack (CVE-2014-3566), Padding Oracle on Downgraded Legacy Encryption. This led to the development of TLS, or Transport Layer Security. TLS uses a stronger encryption known as AES which is less susceptible to brute-force attacks. TLS also uses a more secure method of authentication by using a combination of symmetric and asymmetric encryption to ensure privacy. The TLS handshake was redesigned to be more secure and efficient, reducing the complexity and vulnerability of the initial connection setup compared to SSL (Thurman, 2026).

Smaller companies need to maintain a strong computer incident response team, or CIRT, the same as larger companies do. Because they have limited staff, they often do not create a specialized team but rather choose a team that will come together in the event of a threat. By using clear incident response plans and having continuous training of their teams they can be highly effective in stopping incoming threats and viruses. They should first decide what vulnerabilities exist in their company and what types of targets may attackers want to obtain. Next, they should decide what tools they have or need to have in order to mitigate threats as they occur to protect those assets. Then by documenting the plan in an easy-to-understand method that can be quickly implemented when an attack occurs the CIRT will be effective in neutralizing the threat.

References:

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